

Subhan Shaikh

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SKILLS

- **Languages:** Python, JavaScript, TypeScript, Java, C, SQL.
- **AI / ML & Backend:** RAG, Multi-agent systems & Orchestration, MCP Servers, LLM fine-tuning, Model Evaluation, Inference Optimization, LangChain, LangGraph, Autogen, Hugging Face, Vector databases, REST APIs, PostgreSQL, Supabase, Docker, GCP, Vercel, Git, CI/CD.

WORK EXPERIENCE

Gogentic.ai

June 2025 – Nov. 2025

Applied AI Engineer

Remote

- Delivered a fully functional AI-driven backend MVP in 8 weeks by leading the AI team and establishing efficient backend workflows.
- Reduced AI infrastructure expenses by **80%** by transitioning from third-party APIs to open-source models, optimizing operational spending.
- Boosted backend service performance by **50%** through custom workflows and strategic resource management to enhance reliability and efficiency.

UC Santa Cruz - AIEA Lab

June 2025 – March 2026

Graduate Researcher

Santa Cruz, CA

- Researched advanced Retrieval-Augmented Generation techniques to build a CS-department specific chatbot over course and program data, under Professor Leilani Gilpin's AI Explainability & Accountability Lab. Explored Prolog-based symbolic reasoning as part of broader research into LLM explainability.

IBM

Sep 2022 – Nov 2022

AI Intern

Remote

- Built a sign-language recognition system end to end on the ASL dataset achieving 90% classification accuracy. Delivered a working prototype, documented findings for a stakeholder presentation.

PROJECTS

RepairMate : AI-Driven Home Device Diagnostics | Autogen, OpenCV, Whisper, VLMs

- Architected a multi-modal, multi-agent system that diagnoses household device faults from user-submitted video, fusing frame extraction, transcription, and Vision-Language model (VLM) analysis into detailed and structured technician summaries.
- Automated prompt engineering with **GEPA**, a reflection-based optimizer, jointly tuning frame-analysis and aggregation prompts against BERTScore to improve terminology precision and diagnostic grounding. Applied **model quantization** to **cut model size** and **boost inference speed** for real-time on-device performance.

Critique-Enhanced RAG: Self-Refining Retrieval-Augmented Generation | [GitHub](#) | Python, OpenAI API

- Built a self-critiquing RAG system that scores its own answers across four quality dimensions and conditionally refines those below threshold, improving oracle quality score **27%** over a vanilla baseline on 150 SQuAD dataset questions.
- Benchmarked three architectures (Vanilla, LLM reranker, Critique-Refinement) on quality, cost, and latency, showing the critique loop refined 56% of answers and matched its gains at 2× cost while LLM reranking cost 6× for no measurable improvement.

EDUCATION

University of California, Santa Cruz

June 2026

Master's in Computer Science (GPA: 4.0)

Santa Cruz, CA

University of Mumbai

June 2024

Bachelor's in Computer Science

Mumbai, India